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Understanding Injury Patterns and Predictors in Pickleball Players: A Nationwide Study of 1,758 Participants

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Abstract

Background Despite pickleball's rapid growth in the United States, research regarding the patterns and predictors of injuries remain sparse.

Objectives To describe the prevalence and characteristics of injuries, including time-loss (stopping play for at least a day) and non-time-loss injuries, and evaluate the predictors of injuries in pickleball players.

Methods A cross-sectional study was conducted. Pickleball players ≥ 18 years of age across the United States, who participated in pickleball at least once a month, were invited to take a pre-validated survey. The primary outcome was self-reported all-complaint injuries, including any physical complaints in the past 12 months.

Results A total of 1,758 participants (mean age: 62.7 ± 13.0 years) were included in the final analysis. The 12-month prevalence of all-complaint injuries was 68.5% (95% CI: 66.3–70.7%), with time-loss injuries at 40.8% (95% CI: 38.5–43.1%) and non-time-loss injuries at 51.2% (95% CI: 49.4–54.1%). The point prevalence of pain/ongoing injuries was 35.9% (95% CI: 33.1–38.7%). The knee reported the highest injury prevalence (29.1%) followed by combined lower extremity regions of thigh, leg and foot (26.9%), shoulder (22.2%), back (19.9%) and elbow (18.4%). The top “most serious” injury types were overuse/chronic conditions (35.3%), joint/ligament sprains (23.8%), and muscle strains/pulls (20.7%). Based on a multivariable logistic regression, significant predictors of injury included male sex (OR: 1.33, 95%CI: 1.07–1.65, $p=0.011$), higher frequency of weekly play (OR: 1.45, 95%CI: 1.15–1.84, $p=0.002$), fewer years (< 5 years) of play experience (OR: 1.50, 95%CI: 1.19–1.90, $p=0.001$), low/moderate perception of injury prevention importance (OR: 2.02; 95%CI: 1.52–2.67, $p<0.001$), and age categories ranging from 33 to 77 years (ORs ranging from 1.83 to 3.11, $p\leq 0.009$). Neither increased duration of play nor higher body mass index significantly increased the odds of injury.

Conclusions Injuries are common among pickleball players, with 69% experiencing at least one all-complaint injury annually, including two in five sustaining injuries that halt play and one in three continuing to play despite pain. These findings underscore the need for tailored injury prevention strategies to optimize the health benefits of pickleball. Identified predictors will inform future injury prevention initiatives in pickleball.

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Key Points

- Pickleball is the fastest growing sport in the United States, but research regarding the patterns and predictors of injuries remain sparse.
- Injuries are common among pickleball players, with 69% experiencing time-loss and/or non-time-loss injuries annually and one in three continuing to play despite pain.
- Potential injury predictors in pickleball are male sex, higher frequency of weekly play, fewer years (<5 years) of play experience, lower perception of prevention importance and age category range of 33–77 years.
- This study provides baseline information for future injury prevention research and initiatives for sports medicine clinicians to protect the health of players and maximize the physical and mental health benefits of pickleball among players.

Keywords Pickleball injuries, Risk factors, Epidemiology, Prevention

Introduction

Pickleball is America's fastest growing sport with 48.3 million (about 19% of United States population) reported players in 2023 [1]. The game of pickleball, previously associated with older adults and mostly played in retirement villages, has become a household name across the United States and is now played across the lifespan. There are several health benefits associated with pickleball, ranging from physical to mental health benefits [2]. Pickleball is played as singles or doubles on a court that is 20 feet wide by 44 feet long, divided in half by a 34-inch net. Players use a paddle to hit a perforated plastic ball over the net. Because of its smaller court size and slower ball speed, pickleball is often viewed as a less physically demanding racquet sport than tennis or badminton for example. Nevertheless, pickleball requires a significant amount of physical activity including running (acceleration and deceleration), jumping, and sudden change in movement direction. All of these multidirectional movements subject the body to axial and torsional stress and increase the risk of injury [3].

Pickleball is considered a combination of tennis, ping pong and badminton. As a result, the biomechanical forces the body is put through while playing pickleball are complex, differing from those involved in other racquet sports. Pickleball players must constantly shift their weight and change direction in response to the ball's speed, spin and trajectory. These motions require quick, coordinated movements from the entire body to maintain balance and have an adequate reaction time. The unique paddle design also places additional stress on the wrist and elbow joints, which can lead to overuse injuries over time [4]. Since pickleball can be considered a less physically demanding sport compared to other racquet sports, players may underestimate the potential for injury and therefore neglect a proper warm-up or other evidence-informed injury prevention routines.

Despite being the fastest growing sports in the United States, there is limited research information regarding the patterns and characteristics of injuries

sustained by pickleball players. A 2023 scoping review [5] including four original articles revealed that current pickleball injury epidemiology is mostly based on administrative record data from emergency departments using the National Electronic Injury Surveillance System [6–9]. A major limitation with emergency record data is that they do not capture non-medical attention injuries (i.e., injuries that do not result in a player seeking medical care), including pain or overuse injuries [10]. The single cross-sectional study [6] included in the review had a sample size of 129 adults (≥ 50 years) in the state of Georgia, a limited sample size and participant demography that is not representative of the pickleball community in the United States. Moreover, the survey's focus was not primarily on injury epidemiology. Contemporary evidence demonstrates that traditional injury surveillance methods based only on medical attention and/or time loss (i.e., inability to continue to participate in sports) are insensitive to capturing all injuries. This methodology underestimates injury risk, pattern and severity, particularly pain and/or overuse injuries and/or traumatic injuries that do not result in time loss [10, 11]. Self-report methodologies incorporating “all-complaint injuries” (i.e., capturing time-loss and non-time-loss, including traumatic, medical attention, overuse injuries and/or pain) have been shown to be a more accurate surveillance method to fully capture injury burden, especially in sports projected to have a high risk of overuse injuries [10, 12–14].

An important initial step towards mitigating the risk of pickleball injuries is to understand injury patterns and predictors [4, 15]. For a sport like pickleball, measures of prevalence are an appropriate metric given the high potential for overuse conditions and chronic pain. A better understanding of injury occurrence and associated factors using a self-report all-complaint injury methodology could help with identifying players at a higher predisposition for injuries, towards developing and implementing tailored pickleball-specific injury prevention interventions for players across the United States and globally. In the current study we: (i) Describe

the 12-month and point prevalence of all-complaint injuries, time-loss injuries, and non-time-loss injuries, (ii) Describe the characteristics of injuries and (iii) Evaluate the predictors of injuries in active pickleball players.

Methods

Study Design and Setting

This study is the foundational component of an ongoing project, the Surveillance in Pickleball players to reduce INjury burden (SPIN), a hybrid clinical and implementation research project with an overarching objective of developing pickleball-specific injury prevention interventions. A national online survey was conducted on active pickleball players across the United States to examine injury patterns and associated predictors.

Participants

All pickleball players who engaged in any form of play (from recreational to competitive) at least once a month, reside in the United States and are ≥ 18 years old were eligible to participate in this study. The once-a-month criterion was selected to eliminate any potential bias of players not having adequate pickleball exposure through the past 12 months. We targeted players on a spectrum of skill levels and experiences. Enrollments occurred across several settings nationwide, but primarily through social media advertisements (Instagram, Facebook [Meta Platforms, Menlo Park, CA], X [X-Corp, Bastrop, Texas]) and partnership with DUPR (Dynamic Universal Pickleball Rating, Ft. Lauderdale, FL). Other recruitment routes included use of recruitment flyers containing survey link or QR code distributed at pickleball clubs, tournaments, organizations, organized community settings, colleges, and local gyms. Individuals were excluded from the study if they reported an age of < 18 years or indicated residence outside of the United States. The study protocol was reviewed and approved by the Saint Louis University Institutional Review Board before commencement (#33859). All participants reviewed a recruitment statement detailing the purpose and procedures of the study before completing the anonymous SPIN survey. Using the SurveyMonkey (San Mateo, California) online sample size calculator, an estimated sample size of 1,985 pickleball players was set a priori based on a 2.2% margin of error and a confidence level of 95%, considering a 14% population proportion of pickleball players in the United States [16].

The SPIN Survey and Variables of Interest

Player demographics/characteristics, potential injury predictors, injury outcomes variables and all other variables of interest were collected using a semi-structured questionnaire (Supplementary File) adapted from previous studies with similar focus and with an integration of

the Oslo Sports Trauma Research Center Overuse Injury Questionnaire [6, 12, 17, 18]. Specific questions included an array of concepts comprising demographics, medical history, years of experience, frequency of play, level of play, injury history (12-month history) and current injury characteristics e.g., injury type (sprains, fractures, concussions), medical attention/time loss injury details, body parts affected etc. In addition, information was collected related to participants' context to inform future injury prevention intervention development, implementation and evaluation. We used closed- and open-ended questions regarding current knowledge of pickleball injury risk, current injury prevention practices, and perceived barriers to and facilitators of injury prevention in pickleball (findings to be presented in a separate study). The SPIN survey was administered online using QualtricsXM (Provo, Utah) and it took seven to ten minutes on average to complete.

Before study commencement, the SPIN survey was pilot tested to ensure face and content validity. Following an initial draft [6, 17] and in consultation with pickleball stakeholders (players across levels and experienced coaches), the SPIN survey proceeded through several iterations of drafts based on reviews and revisions by a team of clinician-scientists (including the authors) with several years of experience (range: 3–26 years) and expertise in sports injury prevention, epidemiology, sports physical therapy/medicine, orthopedic surgery, qualitative research methods, dissemination and implementation research and survey tool development. The majority (6 of 9) of the author panel are also active pickleball players (playing at least once a month as a recreational or competitive player). The process of content validation was followed by a face validation among 12 local pickleball players. The survey was further revised for clarity and relevance based on the feedback obtained.

Injury Outcomes and Potential Predictors

The primary injury outcome was a self-report of at least one all-complaint injury, i.e., all forms of physical complaints regardless of loss of subsequent pickleball participation, in the past 12 months. The primary question used to capture all-complaint injuries was: "Have you experienced any physical complaints, discomfort (e.g., pain, ache, stiffness) or injuries related to pickleball in the past 12 months?" This question was followed by additional questions used to classify all-complaint injuries into either time-loss or non-time-loss. Time-loss injury was defined in the survey as: "A physical complaint, discomfort or problem that resulted in at least 1 day of missed participation from pickleball and/or another forms of exercise" and non-time-loss defined as: "A physical complaint, discomfort or problem that did not stop you from playing pickleball and/or any other forms of exercise". An

additional injury outcome was point (current) prevalence of pain/overuse injuries, i.e., self-reported current pain, aches or discomfort associated with pickleball. Lastly, players were asked to select from a list of injury types that most closely describe their “most serious physical complaints or injuries in the past 12 months.”

Potential predictors of injury included age, sex, body mass index (BMI), years of pickleball playing experience, weekly hours of play, level of play, average play time, prior knowledge of sports injury prevention, perceived importance of injury prevention in pickleball and regular participation in other sports. A priori hypotheses were made for all predictors of interest. The hypotheses informed the thresholds for classifying exposed vs. non-exposed players. For example, we hypothesized that (i) male participants will have higher odds of all-complaint injuries compared with female players; (ii) participants playing pickleball 3x or more weekly will have higher odds of all-complaint injuries compared with those playing 2x or less; (iii) injury risk will vary by age groups (iv) injury risk will vary based on playing years of experience.

Statistical Analysis

Player characteristics and other variables of interest were reported separately for male and female players using descriptive statistics. The 12-month injury prevalence with 95% confidence interval (CIs) for all-complaint injuries, time-loss and non-time-loss injuries were calculated for all players and by sex. Point prevalence with 95% CI was also calculated. Injury distribution was described by regions of the United States and sex-specific injury proportions were described in terms of body location and type of injury. Specific predictor variables of interest such as sex, age (in categories of 15-year intervals to isolate differences across age groups), BMI, years of pickleball playing experience, weekly hours of play, level of play, average play time, prior knowledge of sports injury prevention and perceived importance of injury prevention (all treated as categorical data except BMI) were examined in a multivariable logistic regression model while evaluating variables for potential confounding effect in the multivariable regression model. Adjusted odds ratio (ORs) were reported with 95% CI. An alpha level of 0.05 was used as a threshold of statistical significance for the inferential statistics of regression.

Results

Player Characteristics

A total of 2,054 pickleball players participated in the SPIN Survey. A total of 1,758 participants (mean age: 62.7 ± 13.0 years; age range 18–102 years) had complete injury data for final analysis with survey margin of error of 2.35% (0.15 higher than our estimated sample size, using the same confidence level). The specific reasons for

participant exclusion were: (i) not a resident in the United States, $n = 56$ (2.7%) (ii) under-aged (< 18 years), $n = 3$ (0.1%) and (iii) incomplete data, i.e., no injury outcomes, $n = 265$ (12.9%). Most of the players were white (94.9%) and recreational players (98.8%). The majority (62.6%) of the players were retired and many (68.6%) played pickleball 3x or more weekly. Details of overall and sex-specific player characteristics are presented in Table 1.

Injury Prevalence and Characteristics

The overall 12-month injury prevalence of all-complaint injury was 68.5% (95%CI: 66.3–70.7%), with a higher prevalence in male (70.6%) vs. female (66.9%) players. The overall 12-month prevalence of time-loss injuries was 40.8% (95%CI: 38.5–43.1%), non-time-loss injuries was 51.2% (95%CI: 49.4–54.1%) and point prevalence of pain was 35.9% (95%CI: 33.1–38.7%). The sex-specific details of injury outcomes are presented in Table 2. The prevalence of all-complaint injuries varied across age groups – from 48.4% in 18–32-year-olds to 77.3% in 48–62-year-olds (Table 3) and across regions of the United States – from 64% in the Southwest region to 77% in the West region (Fig. 1).

The prevalence of all-complaint injuries by body part and sex is presented in Table 4; Fig. 2. Overall, the knee was the most commonly reported injury site (29.1%, 95% CI: 27.0–31.3), followed by the combination of the thigh, leg, or foot (26.9%, 95% CI: 24.8–29.0) and the shoulder (22.2%, 95% CI: 20.3–24.2). When stratified by sex, males most frequently reported injuries to the thigh, leg, or foot (31.8%, 95% CI: 28.6–35.2) and knee (30.5%, 95% CI: 27.3–33.9). Females most commonly reported knee (28.0%, 95% CI: 25.2–30.9) and shoulder injuries (20.1%, 95% CI: 17.6–22.7). Other frequently reported injury locations included the back (19.9%), elbow (18.4%), and hip (14.2%), with similar prevalence between sexes. Less commonly reported were injuries to the wrist, forearm/arm/hand, and head/neck, all below 11%.

A comparable distribution was reported for non-time-loss (53.7%) vs. time-loss injuries for the “most serious” injuries among players. The overall top injury types based on the “most serious” injuries players had over the past 12 months were overuse or chronic conditions (35.3%), joint/ligament sprains (23.8%) and muscle strains/pulls (20.7%). Concussion (0.5%) and dislocations (0.3%) were the least reported injuries (Table 5).

Injury Predictors

The predictor variables with statistically significant associations with all-complaint injury in a multivariable logistic regression model included sex, age, frequency of weekly pickleball play, pickleball play experience and perception of sports injury prevention importance (Table 6). Specifically, the odds of an all-complaint injury was 33%

Table 1 Characteristics of the pickleball players

	Overall (N = 1758) n (%)	Males (N = 786) n (%)	Females (N = 972) n (%)
Age Group (Years)			
≥65	985 (56.0)	475 (60.4)	510 (52.5)
18–64	773 (44.0)	311 (39.6)	462 (47.5)
State Region			
Midwest	758 (43.1)	346 (44.0)	412 (42.4)
Southeast	412 (23.4)	199 (25.3)	213 (21.9)
Northeast	245 (14.0)	94 (12.0)	151 (15.5)
West	206 (11.7)	80 (10.2)	126 (13.0)
Southwest	137 (7.8)	67 (8.5)	70 (7.2)
Race			
White	1669 (94.9)	745 (94.8)	924 (95.1)
Asian	54 (3.1)	26 (3.3)	28 (2.9)
Black or African American	19 (1.1)	12 (1.5)	7 (0.7)
American Indian or Alaska Native	9 (0.5)	2 (0.3)	7 (0.7)
Native Hawaiian or Pacific Islander	7 (0.4)	1 (0.1)	6 (0.6)
Weekly Pickleball Play			
3x or more	1206 (68.6)	534 (67.9)	672 (69.1)
Up to 2x	552 (31.4)	252 (32.1)	300 (30.9)
Weekly Play Time			
Up to 2 h	908 (52.4)	383 (48.7)	525 (54.0)
More than 2 h	825 (47.6)	403 (51.3)	422 (46.0)
Playing Experience (Approximate Years)			
<5	1275 (72.5)	548 (69.7)	727 (74.8)
≥5	483 (27.5)	238 (30.3)	245 (25.2)
Level of Play			
Recreational	1766 (98.8)	770 (98.0)	996 (99.4)
Advanced/Professional	22 (1.2)	16 (2.0)	6 (0.6)
DUPR Rating			
No	1149 (65.4)	485 (61.7)	664 (68.3)
Yes	609 (34.6)	301 (38.3)	308 (31.7)
DUPR Rating Level			
3.0–3.9 (Intermediate)	441 (72.4)	209 (69.4)	232 (75.3)
≥4.0 (Advanced/Elite)	126 (20.7)	84 (27.9)	42 (13.7)
2.5–2.9 (Beginner)	42 (6.9)	8 (2.7)	34 (11.0)
MSK Sports Injury Prevention Training			
No	1355 (77.1)	611 (77.7)	744 (76.5)
Yes	403 (22.9)	175 (22.3)	228 (23.5)
Concussion Prevention Training			
No	1392 (79.2)	624 (79.4)	768 (79.0)
Yes	366 (20.8)	162 (20.6)	204 (21.0)
Perceived Sports Injury Prevention Knowledge (Out of 10)			
Low/Moderate (1–7)	1410 (80.2)	649 (82.6)	761 (78.3)
High (8–10)	348 (19.8)	137 (17.4)	211 (21.7)
Perceived Sports Injury Prevention Importance (Out of 10)			
High (8–10)	1473 (83.8)	621 (79.0)	852 (87.7)
Low/Moderate (1–7)	285 (16.2)	165 (21.0)	120 (12.4)
Employment Status*			
Retired	1141 (62.6)	509 (62.7)	632 (62.5)
Employed for wages	467 (25.6)	210 (25.9)	257 (25.4)
Self-employed	122 (6.7)	61 (7.5)	61 (6.0)
Student	44 (2.4)	22 (2.7)	22 (2.2)

Table 1 (continued)

	Overall (N= 1758) n (%)	Males (N= 786) n (%)	Females (N= 972) n (%)
Homemaker	27 (1.5)	2 (0.2)	25 (2.5)
Out of work	22 (1.2)	8 (1.0)	14 (1.4)

*Multiple responses allowed

DUPR: Dynamic Universal Pickleball Rating; MSK: Musculoskeletal

Table 2 Prevalence of Injuries in Pickleball Players

Injury Outcome	Overall	Males	Females
All-Complaint Injuries	68.5 (66.3–70.7)	70.6 (67.3–73.8)	66.9 (63.8–69.8)
Time-Loss Injuries	40.8 (38.5–43.1)	40.8 (37.4–44.4)	40.7 (37.6–43.9)
Non-Time-Loss Injuries	51.2 (49.4–54.1)	54.8 (51.3–58.4)	49.3 (46.1–52.5)
Current (Ongoing) Pain	35.9 (33.1–38.7)	35.2 (31.2–39.4)	36.4 (32.6–40.4)

Values are % (95% CI) based on a 12-month injury/pain history except for current pain

Table 3 Prevalence of All-Complaint Injuries by Age-Group (15-Year Intervals)

Age (years)	# of players in age group	# of player reporting injuries	% (95% CI) of player reporting injuries
18–32	93	45	48.4 (37.9–59.0)
33–47	103	79	76.7 (67.3–84.5)
48–62	428	331	77.3 (73.1–81.2)
63–77	1,047	706	67.4 (64.5–70.3)
≥ 78	87	44	50.6 (39.6–61.5)

higher in males compared to females (OR: 1.33, 95%CI: 1.07–1.65; $p=0.011$). Players aged 33–77 years had a significantly higher likelihood of injuries compared to the youngest group aged 18–32 years ($p<0.01$); however, no significant differences were observed for players aged 78 years or older (OR=1.01, 95% CI: 0.55–1.87, $p=0.971$). Players who reported playing pickleball three or more times per week had a 45% higher likelihood of sustaining injuries compared to those who played up to two times per week (OR=1.45, 95% CI: 1.15–1.84, $p=0.002$); however, the effect of playing frequency on injury prevalence varied across categories of years of pickleball experience (Fig. 3). Players with less than five years of pickleball experience were 50% more likely to report injuries compared to players with five or more years of experience and players with a low/moderate perception of importance for sports injury prevention were two times more likely to report injuries compared to those with a high perception of importance. Conversely, average play duration per session, sports injury prevention knowledge, and BMI were not statistically significant predictors of injury, with p -values of 0.148, 0.072, and 0.088, respectively.

* Statistical significance at $\alpha \leq 0.05$. OR: odds ratio; CI: confidence interval. In addition to all the variables included, the final model was adjusted for regular participation in other sports (yes vs. no) and level of play (recreational/amateur vs. advanced/professional); both were not significantly associated with injury outcome and were not confounders in the model.

Discussion

This study is the first and largest observational study to extensively document injury prevalence and predictors in pickleball players. In this study, we examined the 12-month prevalence of injuries based on an all-encompassing injury definition of “all-complaint” to effectively capture all injury types regardless of severity or medical attention (i.e., participants not seeking medical attention). An overall prevalence of 68.5% reported in our study is higher than that reported in a previous survey by Walton-Mouw et al. [6] conducted in 2021, where a 50% prevalence was reported in 129 pickleball players (aged ≥ 50 years) residing in the state of Georgia. The higher prevalence may be related to the comprehensive all-injury type methodology used in our study and may also be explained by the increasing number of players and facilities across the United States. The latest data from the Association of Pickleball Players reported a 24% (from 36.5 million to 48.3 million players) increase in participation between 2022 and 2023 [1]. Historically, as a sport’s popularity increases, there is often a corresponding rise in the number of reported injuries. This is valid for pickleball given the findings from emergency record data showing a sharp increase in the number of injuries reported between 2020 and 2022 [19]. The increased pickleball participation numbers are a possible explanation for the high prevalence of injuries seen in the current study and the increased emergency room injury numbers across the United States.

Regional differences in injury prevalence were observed, with the West region reporting the highest prevalence of all-complaint injuries with a 77% prevalence compared to 64% in the Southwest. According to the latest Association of Pickleball Professionals report [1], Los Angeles and New York were noted as the two top “hotspot” cities with the highest number of pickleball players in the United States. These cities are in the

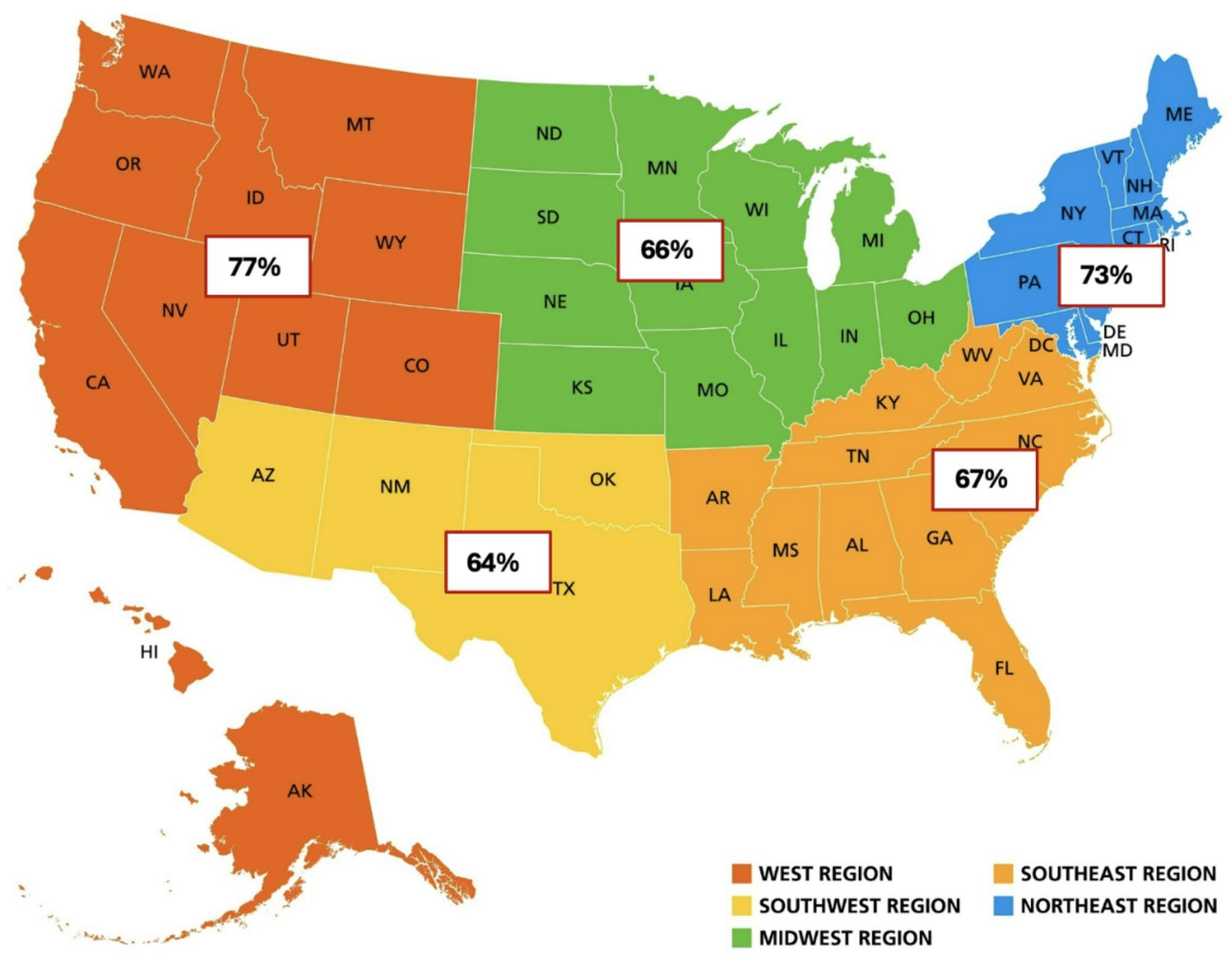


Fig. 1 All-Complaint Injury Prevalence by Regions of the United States

Table 4 All-Complaint Injury Prevalence by Body Part and Sex

	All Players		Males		Females	
	%	95% CI	%	95% CI	%	95% CI
Knee	29.1	27.0–31.3	30.5	27.3–33.9	28.0	25.2–30.9
Thigh, Leg, Foot	26.9	24.8–29.0	31.8	28.6–35.2	22.9	20.3–25.7
Shoulder	22.2	20.3–24.2	24.8	21.8–28.0	20.1	17.6–22.7
Back	19.9	18.0–21.8	20.1	17.4–23.1	19.7	17.2–22.3
Elbow	18.4	16.6–20.3	21.6	18.8–24.7	15.7	13.5–18.2
Hip	14.2	12.6–15.9	12.5	10.2–15.0	15.5	13.3–18.0
Ankle	12.5	11.0–14.2	13.7	11.4–16.3	11.5	9.6–13.7
Wrist	10.1	8.8–11.6	9.8	7.8–12.1	10.4	8.5–12.5
Forearm, Arm, Hand	8.9	7.6–10.3	8.4	6.6–10.6	9.3	7.5–11.3
Head and Neck	8.5	7.3–9.9	9.3	7.4–11.5	7.9	6.3–9.8

regions with the highest injury prevalence: West and Northeast regions, respectively. The higher injury prevalence in these regions may be associated with higher pickleball participation rates. This geographic variation may reflect differences in playing conditions, climate, or regional player demographics and habits.

The findings from this study provide valuable insights into the injury patterns and mechanisms among pickleball players. Overall, non-time-loss injuries – which are often pain or overuse-related – were more common than time-loss injuries, which are typically acute or traumatic. Approximately one in two players reported experiencing non-time-loss injuries, while two in five reported

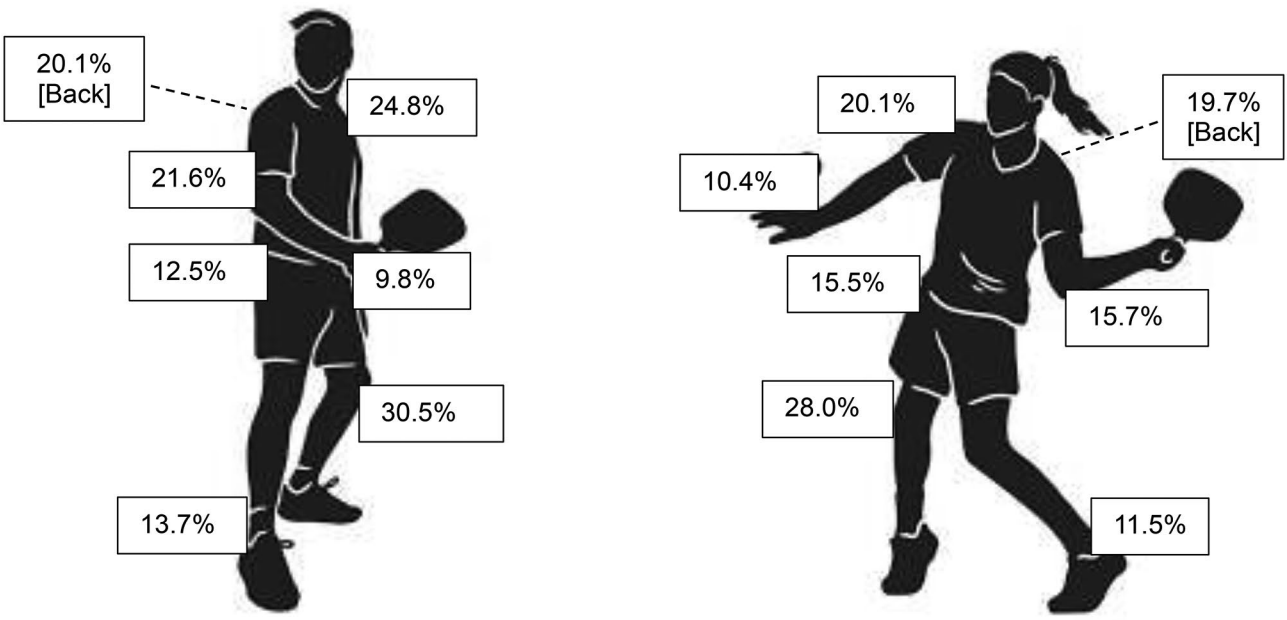


Fig. 2 All-Complaint Injury Prevalence by Joints in Male and Female Pickleball Players

Table 5 Injury Types Based on “Most Serious” Pickleball-Related Injuries among Players

	All Injuries		Males		Females	
	#*	%	#	%	#	%
Overuse or chronic conditions, including tendon injuriestendinopathy	628	35.3	269	32.7	359	37.6
Joint/ligament sprain, joint swelling	424	23.8	214	26.0	210	22
Muscle strain or pulled muscle	368	20.7	209	25.4	159	16.7
Abrasion, blisters, bruise, cuts	162	9.1	60	7.3	102	10.6
Others (not specific to pickleball e.g., arthritis, sciatica, neuroma, burn etc.)	106	6.0	44	5.4	62	6.5
Broken bone, fracture	58	3.3	12	1.4	46	4.8
Sunburn or heat stroke	17	1.0	5	0.6	12	1.3
Concussion/knocked out	9	0.5	6	0.7	3	0.3
Dislocation	6	0.3	4	0.5	2	0.2
Total	1778	100.0	823	100.0	955	100.0

*Multiple responses allowed

time-loss injuries. In addition, one in three players had an ongoing pain/overuse condition. These findings suggest that non-time-loss injuries are a more frequent issue than time-loss injuries and likely an ongoing problem in some players, reflecting the repetitive nature of pickleball movements and its potential cumulative effects on musculoskeletal health. The knee emerged as the most frequently injured body region for both male and female players, highlighting its critical role in pickleball movements that require agility and rapid directional changes [3]. This was closely followed by injuries to the shoulder, back, and elbow, emphasizing the importance of upper body mechanics in the sport. Furthermore, the combined lower extremity regions (thigh, leg, and foot) accounted for a significant proportion of injuries (16.1%), reflecting the high physical demands placed on the lower body [3]. These findings corroborate the findings from emergency

departments across the United States in which the lower extremity was reported as the most injured body region, followed by the upper extremity and the trunk [5, 9]. Injury characteristics differed significantly between sexes for lower extremities, with males reporting higher time-loss injuries (26.4%) overall, compared to females (17.4%). Males reported higher prevalence of muscle strains (males: 25%, females: 17%) while females reported higher prevalence of fractures (males: 1.4%, females: 4.8%), consistent with current literature [5, 20]. Differences in play style, intensity, or biomechanics may explain this and warrant further study for targeted prevention. Common injuries included overuse conditions, joint sprains, and muscle strains/pulls, while concussions and dislocations were rare. Unlike emergency department data where fractures are prevalent, our findings show fractures are infrequent, likely reflecting the settings and

Table 6 Multivariable Regression Model Examining the Predictors of All-Complaint Injuries in Pickleball Players

Variable	OR	95% CI	p-value
Sex			
Females (reference group)	1		
Males	1.33	1.07–1.65	0.011*
Age Group			
18–32 (reference group)	1		
33–47	3.11	1.66–5.82	<0.001*
48–62	3.06	1.88–4.98	<0.001*
63–77	1.83	1.16–2.87	0.009*
≥ 78	1.01	0.55–1.87	0.971
Weekly Pickleball Play			
Up to 2x (reference group)	1		
3x or more	1.45	1.15–1.84	0.002*
Average Pickleball Play Duration Per Session			
Up to 2 h (reference group)	1		
3 h or more	1.17	0.95–1.45	0.148
Pickleball Play Experience			
≥5 years (reference group)	1		
<5 years	1.50	1.19–1.90	0.001*
Sports Injury Prevention Knowledge (Likert Scale)			
High (8–10) (reference group)	1		
Low/Moderate (1–7)	1.27	0.98–1.65	0.072
Sports Injury Prevention Importance (Likert Scale)			
High (8–10) (reference group)	1		
Low/Moderate (1–7)	2.02	1.52–2.67	<0.001*
BMI	1.02	0.99–1.05	0.088*

severity associated with emergency cases and the under-reporting of overuse injuries [5, 8, 9].

The results of this study highlight several factors associated with the likelihood of injuries among pickleball players, providing insights into potential risk modifiers for injury prevention strategies. Overall, males demonstrated a statistically significant 33% higher odds of sustaining injuries compared to females. This sex difference may reflect variations in playing style, intensity, or physiological differences, such as muscle strength and joint flexibility, which have been shown in other sports to influence injury risk [21, 22]. Further investigation into sex-specific injury mechanisms and prevention strategies is warranted. Middle-aged groups (33–47 and 48–62 years) exhibited significantly higher odds of injury (over 3x higher) compared to the youngest age group (18–32 years). The elevated odds in these groups align with the high prevalence of injuries reported for these age ranges (76.7% and 77.3%, respectively). These findings may reflect the combined effects of increased physical engagement, greater competitive intensity, and gradual age-related declines in tissue repair capacity and flexibility [22, 23]. Middle-aged athletes often balance a

high level of engagement in sports with occupational and familial obligations, which may limit their ability to adopt injury prevention strategies, such as adequate warm-up and recovery routines [24]. In contrast, players aged 63 years and older displayed lower injury prevalence odds of injury relative to the middle-aged groups. This reduction, despite a still relatively high injury prevalence, may be due to adaptive play strategies, such as reduced play intensity and improved self-regulation. Older players might prioritize injury prevention measures like adequate warm-ups or focus on less physically demanding gameplay.

Players who participated in pickleball three or more times per week were 45% more likely to experience injuries than those who played up to two times weekly, and less experienced players (<5 years) exhibited significantly greater odds of injury compared to players with five or more years of experience. The interaction between playing experience and frequency indicates that the heightened risk of injury associated with frequent play diminishes as players gain more pickleball experience. This suggests that playing experience may act as a strong protective factor among pickleball players. Novice players may be particularly vulnerable to injuries when playing frequently, highlighting the need for targeted interventions like warm-up and recovery routines and/or skill development programs for newer players. Although there remains limited research to support the health benefit claims linked with regular participation in pickleball, there is growing evidence that regular pickleball participation has significant mental health benefits in middle-aged and older adults [2]. An absolute pickleball participation restriction is therefore not a best practice approach to mitigating pickleball injuries, rather, an evidence-based neuromuscular warm-up exercise program and an extensive recovery protocol including education on progressive pickleball exposure should be considered.

Players with low-to-moderate perceptions of the importance of injury prevention were over twice as likely to sustain injuries as those who rated it highly. Interestingly, low-to-moderate sports injury prevention knowledge showed a trend toward higher injury odds, though this did not reach statistical significance ($p = 0.072$). This highlights the critical role of both knowledge dissemination and fostering an injury-conscious mindset among players. The current study found that neither increased pickleball play duration nor higher BMI significantly increased the odds of injury. While prolonged duration of play is often associated with fatigue-related injuries due to repetitive stress and reduced neuromuscular control, the unique gameplay and pacing of pickleball may allow players to better regulate their exertion levels, potentially reducing injury risks associated with prolonged activity. Additionally, the intermittent nature of pickleball and

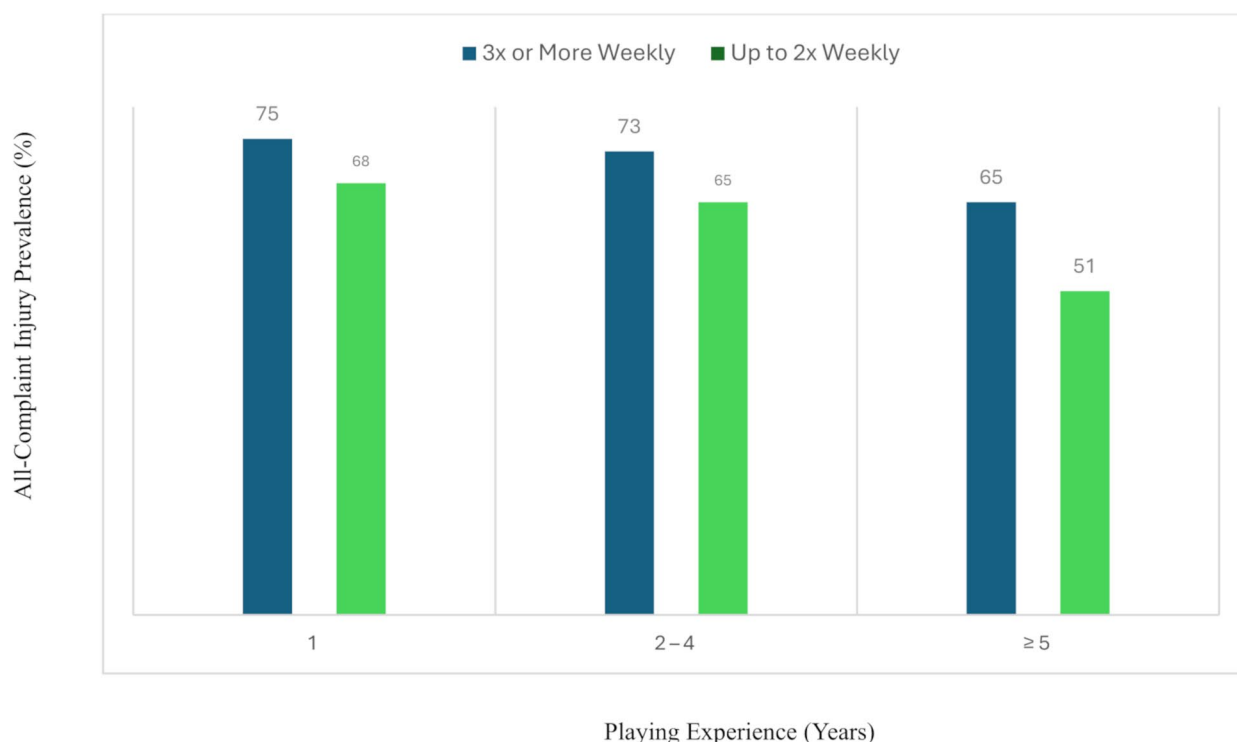


Fig. 3 The Moderation Effect of Playing Experience in the Relationship between Playing Frequency and Injury Prevalence in Pickleball Players

the relatively small court size may reduce physical strain compared to sports requiring continuous high intensity play. Unlike some other sports where higher BMI has been linked to increased joint loading and musculoskeletal strain [25, 26] pickleball allows lower impact movements, potentially mitigating the influence of BMI on injury outcomes.

A limitation of the current study is the lack of diversity in our sample's demographics. For example, the mean age in the study was 68 years and the study sample were mostly white. Given that the national mean age continues to drop and is currently at 35 years [1] there is a need for future studies to intentionally target younger players. Another limitation is the cross-sectional study design. Although the large sample size and broad geographic representation strengthen the findings, the cross-sectional design limits causal inferences. Future research should adopt prospective panel designs (e.g., weekly injury reporting) and explore injury patterns over time, considering factors included in the current study and other factors like biomechanics and equipment. Our findings provide a generalizable baseline for advancing injury prevention in pickleball, emphasizing the need for the development and evaluation of interventions such as neuromuscular warm-ups, recovery protocols, load management, ongoing pain management and injury prevention education through controlled trials.

Conclusion

Injuries are common among pickleball players, with 69% experiencing at least one all-complaint injury annually. Two out of five players may experience an injury that will stop them from continued participation in pickleball and one in three players play with pain. This study provides foundational evidence to advance injury prevention initiatives in pickleball, emphasizing the need for tailored multi-component interventions, including interventions that target neuromuscular deficits in the trunk and extremities, aimed at maximizing the health benefits of pickleball among players.

Abbreviations

SPIN	Surveillance in Pickleball players to reduce INjury burden
CI	confidence interval
OR	odds ratio
BMI	body mass index
DUPR	Dynamic Universal Pickleball Rating

Supplementary Information

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Supplementary Material 1

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Author Contributions

OO conceptualized the study, and all authors contributed to the study design. All authors contributed to data collection and management. OO conducted the data analysis and interpretation. All authors provided critical revisions and methodological guidance. OO, TY and TH drafted the manuscript, while all authors reviewed and approved the final version.

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Data Availability

The data that support the findings of this study are not openly available but are available from the corresponding author upon reasonable request.

Declarations

Consent for Publication

Not applicable.

Conflict of interest

All authors have no conflicts of interests that are directly relevant to the content of this article.

Ethics Approval

The study protocol was reviewed and approved by the Saint Louis University Institutional Review Board before commencement (IRB #33859).

Consent To Participate

All participants reviewed a recruitment statement detailing the purpose and procedures of the study before completing the anonymous SPIN survey.

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